THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. If HCF$(90, 144) = 18$, then LCM$(90, 144)$ is: 1

(a) 720 **(b)** 360 **(c)** 1440 **(d)** 540

2. The prime factorisation of 7429 is: 1

(a) 17×437 **(b)** $13 \times 19 \times 29$ **(c)** $17 \times 19 \times 23$ **(d)** $7 \times 11 \times 97$

3. A quadratic polynomial whose zeroes are 0 and 4 is: 1

(a) $x^2 - 4x$ **(b)** $x^2 + 4$ **(c)** $x^2 + 4x$ **(d)** $x^2 - 4$

4. A quadratic polynomial has sum of zeroes 0 and product -5. Student A writes $x^2 - 5$; Student B writes $x^2 + 5$. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

5. The solution of $x + y = 10$ and $x - y = 2$ is: 1

(a) $(8, 2)$ **(b)** $(5, 5)$ **(c)** $(4, 6)$ **(d)** $(6, 4)$

6. A student states: "The lines $x = 2$ and $y = 2$ are parallel." The correct response is: 1

(a) Correct **(b)** Incorrect — they intersect at $(2, 2)$

(c) They are coincident **(d)** Cannot be determined

7. The discriminant of $x^2 + 4x + 5 = 0$ is: 1

(a) 4 **(b)** −4 **(c)** 0 **(d)** 36

8. If 2 is a root of $x^2 + kx + 12 = 0$, then k is: 1

(a) 6 **(b)** −6 **(c)** −8 **(d)** 8

9. The 15th term of the AP $3, 7, 11, \ldots$ is: 1

(a) 60 **(b)** 59 **(c)** 63 **(d)** 55

10. An AP with $a = -2$ and $d = 0$ is: 1

(a) $-2, -4, -6, \ldots$ **(b)** $0, -2, -4, \ldots$ **(c)** $-2, -2, -2, \ldots$ **(d)** $-2, 0, 2, \ldots$

11. In $\triangle ABC$, $DE \parallel BC$ with $\frac{AD}{DB} = \frac{3}{4}$ and $AE = 6$ cm. Then EC is: 1

(a) 10 cm **(b)** 8 cm **(c)** 4.5 cm **(d)** 6 cm

12. If $\triangle ABC \sim \triangle QRP$ and $\angle A = 50^\circ$, then $\angle Q$ is: 1

(a) 90° **(b)** 130° **(c)** 50° **(d)** 40°

13. If $\cot A = 1$ (A acute), then A is: 1

(a) 45° **(b)** 90° **(c)** 30° **(d)** 60°

14. The length of the tangent from a point 26 cm from the centre of a circle of radius 10 cm is: 1

(a) 16 cm **(b)** 36 cm **(c)** 20 cm **(d)** 24 cm

15. An arc of length 22 cm in a circle of radius 21 cm subtends an angle of: 1

(a) 60° **(b)** 30° **(c)** 120° **(d)** 90°

16. The area of a sector of radius 6 cm and angle 120° is (Use $\pi = 3.14$): 1

(a) 37.68 cm² **(b)** 18.84 cm² **(c)** 75.36 cm² **(d)** 12.56 cm²

17. The volume of a cylinder of radius 7 cm and height 2 cm is: 1

(a) 88 cm³ **(b)** 308 cm³ **(c)** 616 cm³ **(d)** 154 cm³

18. The class mark of the class 30–40 is: 1

(a) 35 **(b)** 10 **(c)** 30 **(d)** 40

19. If the mode is 30 and the median is 26, the mean is: 1

(a) 28 **(b)** 24 **(c)** 32 **(d)** 22

20. A letter is chosen from EDUCATION. The probability of a vowel is: 1

(a) $\frac{4}{9}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{5}{9}$ **(d)** $\frac{1}{3}$

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the HCF and LCM of 40, 36 and 126 by prime factorisation. 2

22. Check whether -2 and 3 are zeroes of $x^2 - x - 6$, and verify the sum of zeroes. 2

23. Find the roots of $2x^2 + x - 4 = 0$ using the quadratic formula. 2

24. $\triangle ABC$ has $\angle A = 60^\circ, \angle B = 80^\circ$ and $\triangle PQR$ has $\angle Q = 80^\circ, \angle R = 40^\circ$. Are they similar? Write the correspondence. 2

25. If the distance between $(2, 3)$ and $(5, k)$ is 5 units, find k. 2

26. In $\triangle ABC$ right-angled at B, $AB = 5$ cm and $AC = 13$ cm. Find $\sin A$ and $\tan C$. 2

27. A tower casts a 30 m shadow when the sun's elevation is 30°. Find its height. 2

28. PA and PB are tangents from P to a circle with centre O, and $\angle APB = 70^\circ$. Find $\angle AOB$ and $\angle OAB$. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $3x + y = 9$ and $x - y = -1$. 3

30. Find the mean by the assumed mean method: 3

Class	10–20	20–30	30–40	40–50	50–60
Frequency	2	4	7	5	2

31. Find the sum of all two-digit odd numbers. 3

32. From an external point P, the tangent PT is 12 cm and $PO = 13$ cm. Find the radius, and the distance from P to the nearest point of the circle on PO. 3

33. A clock's minute hand is 10.5 cm long. Find the area swept by it from 9:00 to 9:40. 3

34. A card is drawn from a deck of 52. Find P(red face card), P(spade or ace) and P(not a diamond). 3

35. From a window 12 m above the ground, the angles of elevation and depression of the top and the foot of a building across the street are 60° and 45°. Find the height of the building. 3

OR

Two points are on opposite sides of a 60 m tower, in line with its foot. The angles of elevation of its top from them are 30° and 60°. Find the distance between the points. 3

36. A student says the median of the data below is 40. Verify. 3

Class	0–20	20–40	40–60	60–80	80–100
Frequency	6	9	10	8	7

OR

The median of the following distribution is 28.5 and $\Sigma f = 60$. Find x and y. 3

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	5	x	20	15	y	5

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A park ABCD has corners A(1, 1), B(7, 1), C(7, 9) and D(1, 9). 5

(i) Find the diagonal AC.

(ii) A fountain is at the centre of the park. Find its position.

(iii) Two lamp posts trisect side AB. Find their positions.

(iv) Verify that the fountain is equidistant from A and C.

OR

C(−1, 2) divides the segment joining A(2, 5) and B(x, y) internally in the ratio 3 : 4. Find B and the length AB. 5

38. Prove that $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$. Verify it for $A = 45^\circ$. 5

OR

Prove that $(1 + \tan A)^2 + (1 - \tan A)^2 = 2\sec^2 A$. Find its value for $A = 60^\circ$. 5

39. A solid is a cylinder of radius 3.5 cm and height 10 cm, with a hemisphere at one end and a cone of height 12 cm at the other. Find 5

(i) its volume

(ii) its total surface area.

OR

Case study: A grain silo is a cylinder of radius 3 m and height 10 m topped by a cone of height 4 m. (Use $\pi = 3.14$) 5

(i) Find the volume of the cylinder.

(ii) Find the volume of the cone.

(iii) Find the total capacity.

(iv) Find the sheet area for the curved surfaces.

40. **Case study:** The weights of 50 students are given. Find 5

(i) the mean

(ii) the median

(iii) the mode.

Weight (kg)	40–45	45–50	50–55	55–60	60–65	65–70
Students	5	8	14	13	6	4

OR

Find the mean age by the step-deviation method and the modal age: 5

Age (yrs)	15–20	20–25	25–30	30–35	35–40	40–45	45–50
Persons	3	8	9	10	3	0	2

END OF QUESTION PAPER **Total: 80 Marks**

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